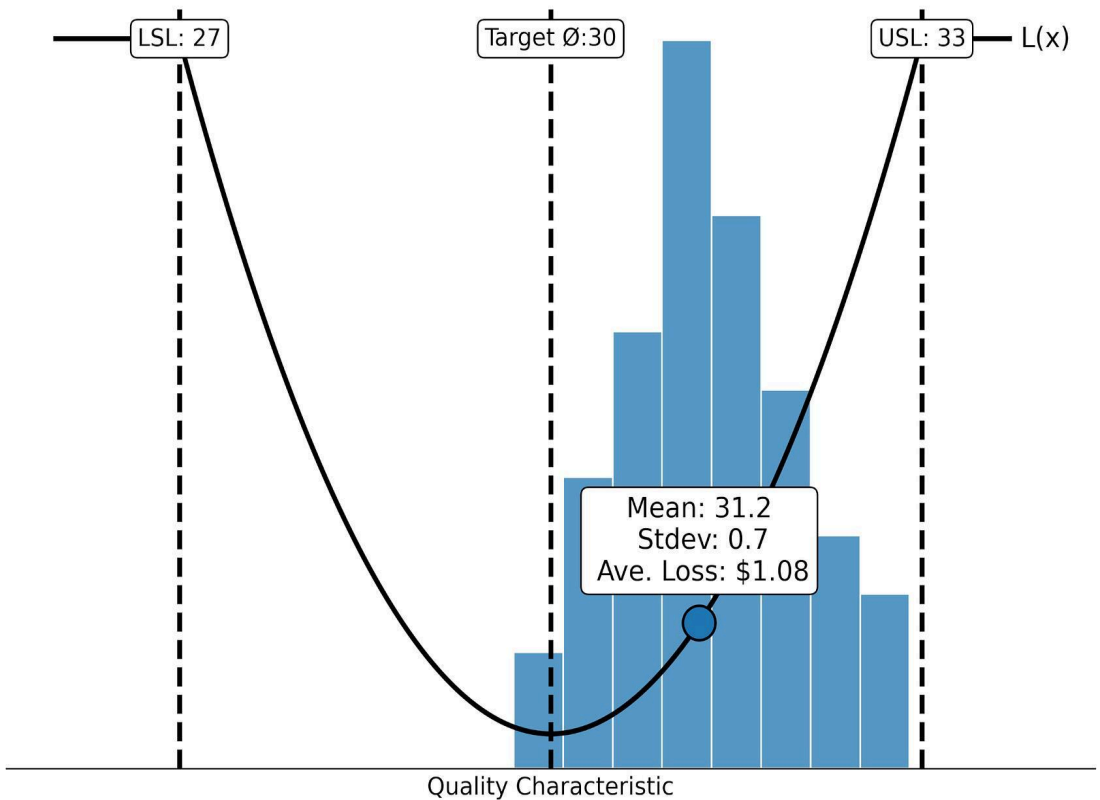


BROKEN QUALITY PRESS PRESENTS

The Taguchi Loss Function

A more accurate model of economic loss due to poor quality



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First edition

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The Taguchi Loss Function

Introduction

Created by the late Dr. Genichi Taguchi, the Taguchi loss function is a response to the adherence to specification (spec) concept of quality that has dominated industry since the start of the Industrial Revolution. Adherence to spec is a binary model of economic loss governed by a square loss function (see Figure 1). Here, anything inside of the specification limits is good and anything outside of them is bad. While this economic model was a step forward when first created, its flaws make it a liability today. The fundamental truth of manufacturing is that *no two parts, assemblies, or products produced by the same process will ever be the same.*¹ Even when parts are in spec, differences will exist because variation always asserts an influence.

¹ This concept is known as the process principle. To learn more about the process principle see the Broken Quality Press article, [*The Process Principle: No two produced by the same process will ever be the same.*](#)

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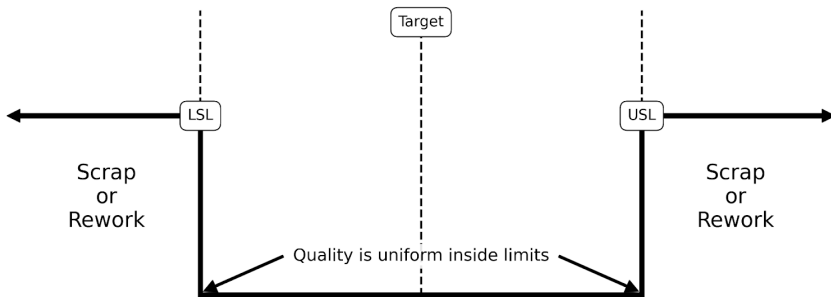


Figure 1. Loss function governing adherence to specification concept of quality.

Recognizing the limitations of adherence to spec, Dr. Taguchi developed the Taguchi loss function. As shown in Figure 2, economic loss under the Taguchi loss function is governed by a parabola. Here, product of the highest quality is produced at the parabolic vertex (target). As a quality or performance characteristic deviates from the target, economic loss increases until the maximum loss is realized at the specification limits. In instances where rework can be performed, a portion of the loss is recovered. In instances where parts must be scrapped, the full economic loss due to poor quality is realized.

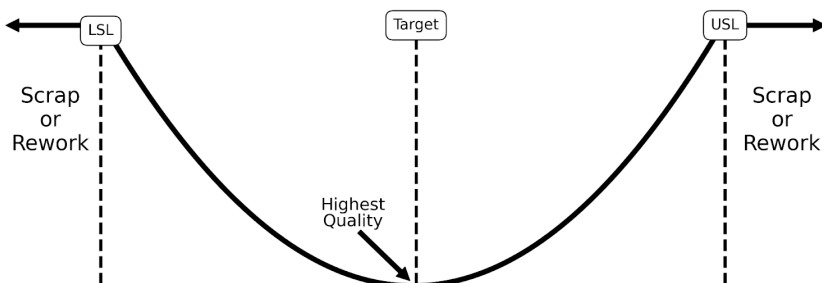


Figure 2. Quadratic loss function governing the continual improvement concept of quality.

The Taguchi loss function is remarkable for how it reflects the realities of manufacturing. It recognizes that variation, in its ubiquity, influences process

outcomes regardless of the specification limits. This promotes a new way of thinking about quality and new goals. Rather than chase “good enough” with adherence to spec, the Taguchi loss function promotes efforts to produce product that is virtually the same. This inevitably led Dr. Taguchi to his definition of world class quality:

On-target with minimum variance.

Operating a process *on-target* requires process knowledge that enables the process mean to be set as close to the target as possible. Operating a process *with minimum variance* requires it to be operated predictably (i.e., in a state of statistical control). The only tool capable of realizing these tasks is the process behavior chart.

An economic model of loss

The Taguchi loss function is not only a conceptually useful model of loss, but also an economic one. That is, it can be used to calculate the financial costs of producing parts off-target with variance. Doing this requires an understanding of the formula that governs the function and its constituent parts.

For an individual quality or performance characteristic (i.e., one measurement from one part), the Taguchi loss function is written as:

$$L(x) = K \cdot (x - T)^2$$

Here, $L(x)$ is the loss function, K is a numeric constant expressed in dollars per unit squared, x is the measured quality or performance characteristic, and T is the target value.

To calculate the value of K , the cost of scrapping a part must be associated with the deviation from the target. This is achieved by connecting the deviation from the target at the specification limits with the cost of scrapping a part,

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C_{scrap} . If we let x_{scrap} represent the measured value of a quality or performance characteristic at the specification limits, the cost of scrap can be written as:

$$C_{scrap} = K \cdot (x_{scrap} - T)^2$$

Here, C_{scrap} is a function of K and the squared deviation of x_{scrap} from the target, T . Rewriting this formula in terms of K yields:

$$K = \frac{C_{scrap}}{(x_{scrap} - T)^2}$$

Thus, the numeric constant K is a function of the cost of scrapping a part and the squared deviation of x_{scrap} from the target, T .

Calculating loss for one off-target measurement

A manufacturing process produces parts with a diameter of 30 units $\pm$ 3 units. This makes the target (T) 30 units, the Upper Specification Limit (USL) 33 units, the Lower Specification Limit (LSL) 27 units, and the tolerance 6 units. The cost of scrap for each part, C_{scrap} , is \$5. Using the Taguchi loss function, what is the economic loss due to poor quality when the measured diameter of a part is 29 units?

To calculate the economic loss associated with a part diameter of 29 units, substitute and solve the Taguchi loss function using the requisite values for K , x , and T . In this example, $x = 29$ and $T = 30$ but K is unknown. Thus, before the loss associated with the part diameter of 29 units can be calculated, the value of K must be determined.

K is a function of the cost of scrap, C_{scrap} , and the squared deviation of x_{scrap} from the target. Here, $C_{scrap} = \$5$, $x_{scrap} = 33$ (or 27, either specification limit will work), and $T = 30$. Substituting these values into the formula for K yields a value of \$0.56.

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$$\begin{aligned} K &= \frac{C_{\text{scrap}}}{(x_{\text{scrap}} - T)^2} \\ &= \frac{5}{(33 - 30)^2} \\ &= \frac{5}{(3)^2} = \frac{5}{9} \\ &= \$0.56 \end{aligned}$$

With the value of K calculated, the economic loss associated with a part diameter of 29 units is calculated as follows:

$$\begin{aligned} L(x) &= K \cdot (x - T)^2 \\ L(29) &= 0.56 \cdot (29 - 30)^2 \\ &= 0.56 \cdot (1)^2 \\ &= 0.56 \cdot 1 \\ &= \$0.56 \end{aligned}$$

Thus, the economic loss associated with a part diameter of 29 units is approximately \$0.56.

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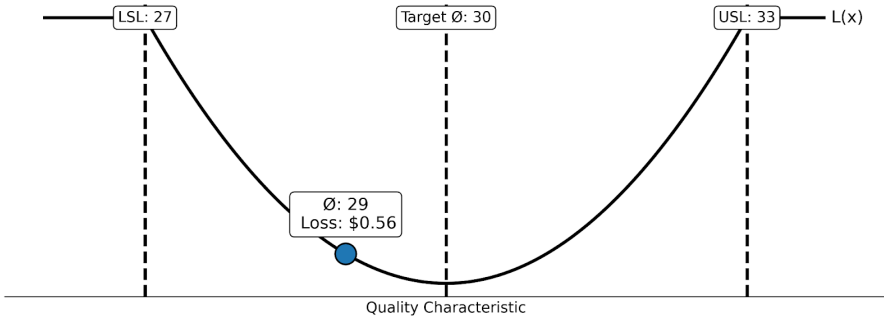


Figure 3. Quadratic loss function with loss associated with a diameter of 29 units.

Calculating loss for a distribution of measurements

Although $L(x)$ can be used to estimate the loss in dollars for a particular value of x , it does not take into account the reality that most manufacturing processes do not typically produce a single part. The purpose and point of manufacturing is to produce multiples of the same part that are, for all intents and purposes, the same. To determine the likelihood that a particular value for a quality or performance characteristic will occur, the distribution of the values of X must be considered.

Assume the previously discussed manufacturing process produced 100 parts. Each of these parts will have a unique quality characteristic, x , and an associated loss that can be described as $L(x)$. By summing the individual losses for the 100 parts, an average loss per unit of production can be obtained. This is expressed by the equation:

$$E\{L(x)\} = K \cdot [(\bar{X} - T)^2 + \sigma^2]$$

Here, $E\{L(x)\}$ is the expected average economic loss per unit, K is again a numeric constant expressed in dollars per unit squared, $\bar{X}$ is the mean of the distribution, T is the target, and σ is the standard deviation statistic.

For the run of 100 units, K is still \$0.56 and the target is still 30 units. The

mean of the distribution is 31.2 units and the standard deviation is 0.7 units. Substituting these values into the expected loss formula yields an average expected loss per unit of \$1.08. This means that if the process continues to operate with a mean of 31.2 units and standard deviation of 0.7 units, the average economic loss per unit will be \$1.08.

$$\begin{aligned}
 E\{L(x)\} &= K \cdot [(\bar{X} - T)^2 + \sigma^2] \\
 &= 0.56 \cdot [(31.2 - 30)^2 + 0.7^2] \\
 &= 0.56 \cdot [(1.2)^2 + 0.7^2] \\
 &= 0.56 \cdot [1.44 + 0.49] \\
 &= 0.56 \cdot 1.93 \\
 &= \$1.08
 \end{aligned}$$

To reduce the average loss per unit, the deviation of the mean from the target and the variation around the mean must be minimized. That is, the process must be operated *on-target with minimum variance*. While opinions abound on how to achieve this task, the only tool capable of realizing it is the process behavior chart.

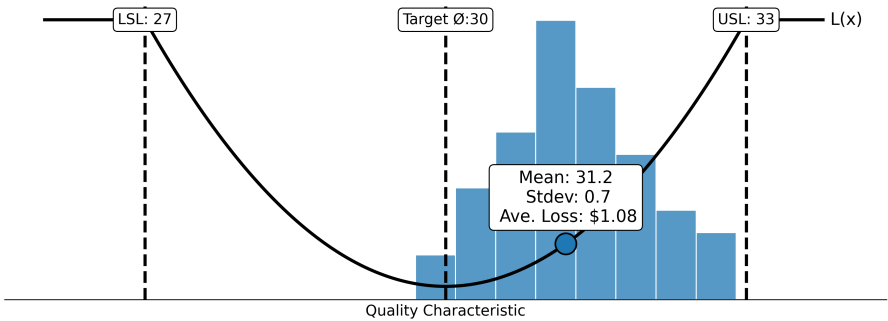


Figure 4. Quadratic loss function with a histogram of 100 part measurements.

Summary

In contrast to the adherence to specification concept of quality, the Taguchi loss function is a high-fidelity model of loss. It trades the binary world of good and bad for one that is based on a quadratic continuum. In doing so, the Taguchi loss function reflects the realities of manufacturing. It recognizes that, whether we like it or not, the measured quality and performance characteristics of parts, assemblies, and products will never be the same (the process principle). Variation always asserts its influence.

The model of loss reflected in the Taguchi loss function is not just theoretical but economic. Given a single measurement or a distribution of measurements, an economic cost can be associated with loss due to poor quality. This connects the world and work of engineers to the world and work of business. It assigns a financial cost to inaction and a financial benefit to operating *on-target with minimum variance*. It promotes a new way of thinking about processes and systems and encourages efforts to continually improve instead of settling for “good enough.”